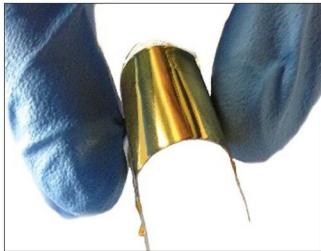


TECHNOLOGY AT YOUR SERVICE

Skin based device can now convert muscle movements into electricity

A team of researchers from National University of Singapore, Singapore, has developed a new method to charge wearable gadgets. The team has developed a tiny new device that can generate electricity from friction produced by human skin. This device generates electricity using tribo-electric effect in skin.



The postage-stamp-sized device uses static electricity to convert mechanical energy into electrical energy

Skin is a great material for tribo-electric effect. This stamp-sized device takes electrons from the skin. Constant frictional movement in skin donates electrons to this device, which makes the device positively charged. The negative layer of the device is made up of flexible silicon rubber, which has thousands of tiny pillar-like structures on it. These structures increase the surface area in contact with the skin. The rubber layer contains 50nm-thick gold film, which acts as an electrode.

Test results show that, if you tap the device with fingers, it produces 90V of current and 0.8MW of power. Researchers tested this device by attaching nano pillar side of the device on a human throat and forearm. The amount of electricity generated from one tap is good enough to light up 12 standard LED lamps. There has been a lot of research in this area and there are some tribo-electric generators already available in the market, but this new product is flexible and very small in size. The team is working on different sizes that can be attached to the human body.

The research team is being led by Chengkuo Lee, a professor of electrical and computer engineering. The device was demonstrated at IEEE MEMS 2015 conference, where a graduate student, Lokesh Dhakar, explained the tribo-electric effect of the skin.

Now turn your regular eyewear into a sunglass

You can experience a new kind of eyewear, which can be transformed into eyeshades whenever you wish. A new kind of lens will make it possible very soon. The lens can convert your regular eyeglasses to shaded ones, and back again, as and when you wish.

The new lens is capable of converting your regular glasses to dark eyeshades within a few seconds and then the glasses can become regular once again in response to a small electrical charge. The charge can be controlled by the user. Lenses have a variety of colours too that can match a wide range of hues.

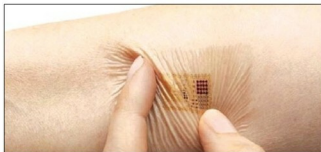
The research is led by Anna Osterholm in John Reynolds' group at the Georgia Institute of Technology, the USA. The study has also pointed out that existing transitional lenses are not made as per users' requirements. For instance, if a user is driving or wearing a baseball cap, then the available lenses usually stay clear and do not switch to the darker mode even in bright sunlight.

Available lenses are also not capable of blocking the harshest light, like reflection from snow. The next generation of eyeglasses will solve these limitations and these will benefit pilots and security guards, among others. According to Osterholm, the colour of lenses can also be fine-tuned to match the range of hues in commercial sunglasses.

Google develops skin to test cancer-detection device

Google is working on a nanoparticle based technology to detect cancer and artificial skin to test it on. Since last year, Google has been working on magnetic nanoparticles that would seek out cancer cells in the bloodstream and report back to a wristband.

Nanoparticles employed in this technology are to use light signals to communicate with the wristband through superficial veins in the wrist.



Google is developing artificial human skin to test cancer-detection devices